FORMOSA HA TINH STEEL CORPORATION
ADDRESS: Vung Ang Economic Zone, Ky Long Ward, Ky Anh Town, Ha Tinh Province, Viet Nam
Tel: +84-239-3722-123 Ext: 6821 Fax: +84-239-3722-112

SAFETY DATA SHEET (SDS)

TECHNOLOGY DIVISION
QUALITY PASSED
CHẤT LƯỢNG BAN HÀNH
BỘ PHẬN KỸ THUẬT LUYỆN KIM

1. PRODUCT AND COMPANY IDENTIFICATION:

- Product Name:

Hot Rolled Products include following main types:

a. Hot Rolled Coil & Hot Rolled Coil, Non-Skinpassed
b. Wire Rod & Bar In Coil,
c. Semi-Finished Products, inclusive of Carbon Steel Slab, Bloom, Billet.
Product name may be changed by customers' requirement.

- Company Name:
  FORMOSA HA TINH STEEL CORPORATION
- Address:
  Vung Ang Economic Zone, Ky Long Ward, Ky Anh Town, Ha Tinh Province, Viet Nam

2. HAZARDS IDENTIFICATION

Steel products in the solid state present no fire or explosion hazard.
However, the fumes resulting from welding. The fine particles resulting from grinding may irritate the respiratory organs and eyes. Burns may result from arcing. Moreover, cutting chips or the like may injurious to the skin.

3. COMPOSITION/INFORMATION ON INGREDIENTS

- Common Name (Symbol), CAS[(a)] Number and Weight

Concentrations of elements are expressed as % by weight.

Common Name (Symbol)	CAS NO.	Weight (%)
Base metal: Iron (Fe)	7439-89-6	Main (>95%) [(b)]
Alloy elements: Manganese (Mn)	7439-96-5	< 5% [(b)]

NOTE:
a. CAS: Chemical Abstract Service
b. Composition values vary within the concentration ranges mentioned above depending on the steel type/standard.

4. FIRST AID MEASURES

A. Description of necessary measures

- Inhalation:
  The product is not a likely form of exposure. The product is unlikely can be inhaled in the supplied form.
- Eye contact:
  The product is not a likely form of exposure. However during further processing (welding, grinding, burning, etc.), once contact rinse the eye(s) cautiously with water for several minutes.
- Skin contact:

After using, wash thoroughly after handling. If irritation or rash occurs, Get medical advice or attention.

B. Most important symptoms/effects, acute and delayed

The product is not likely to present an acute or chronic health effect

TECHNOLOGY DIVISION
QUALITY PASSED
CHẤT LƯỢNG BAN HÀNH
BỘ PHẬN KỸ THUẬT LUYỆN KIM

5. FIRE FIGHTING MEASURES

A. Flammability Classification

The product is in the not flammable or auto-ignition state. B. Means of Extinction:

Use fire-fighting equipment depend on the surrounding materials.

C. Hazardous Combustion Products:

Use self-contained respiratory protection. Toxic and irritating vapors may be released if the product melts or burns in a fire.

6. ACCIDENTAL RELEASE MEASURES

Not applicable to the product in solid state.

7. HANDLING AND STORAGE

A. Handling:

All employees should be trained to handle this product safely. Wear personal protective equipment, when handling the product. B. Storage:

The product should be prevented from physical damage, moisture, and direct sunlight.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

A. Exposure Controls

The product does not present inhalation, ingestion or contact hazard.

B. Personal Protection

Dependent on the process to be performed on the product. Each operation must be addressed for suitable equipment.

9. PHYSICAL AND CHEMICAL PROPERTIES

Item	Result
Appearance	Solid, from Metallic gray to reddish brown (oxide)
Odor	Odorless
Odor threshold	Not applicable
pH	Not applicable
Melting Point	~ 1535℃
Initial boiling point and boiling range	Not determined for the product.

Flash point	Not applicable
Evaporation rate	Not applicable
Flammability (solid, gas)	Not flammable
Upper/lower flammability or explosive limits	Not applicable
Vapor pressure	Not applicable
Vapor density	Not applicable
Relative density	~ 7.85 g/cm^3
Solubility	Not soluble in water.
Partition Coefficient n-octanol/water	Not applicable
Decomposition Temperature	Not applicable
Auto-ignition temperature	Not applicable

10. STABILITY AND REACTIVITY

A. Chemical Stability:

The product is stable under normal storage and handling conditions.

B. Conditions Of Reactivity:

The product is stable under normal conditions.

11. TOXICOLOGICAL INFORMATION

No related information is available at present.

TECHNOLOGY DIVISION
QUALITY PASSED
CHẤT LƯỢNG BAN HÀNH
BỘ PHẬN KỸ THUẬT LUYỆN KIM

12. ECOLOGICAL INFORMATION

No related information is available at present.

13. DISPOSAL CONSIDERATIONS

The product and its scraps or rust can be recycled.

14. TRANSPORT INFORMATION

No special transportation classifications. Since the product is heavy, carry and transport the product carefully.

Cover the product with tarpaulin to prevent the products from direct water or humidity.

15. REGULATORY INFORMATION

No particular applicable regulations

16. OTHER INFORMATION

A. Prepared by :

NGUYEN THI THANH TÂM (Section Manager, Metallurgical Specification & Technical Service Dept. of Technology Division)

B. Remarks :

The information in this SDS is based on the best understanding of the safety and health hazard information of this product. It is used to carry out various operations and processing safely of this product. The information in this SDS applies to the specific materials designated herein.

--Established by FHS Technology Division, FORMOSA HA TINH STEEL CORPORATION--

----------FHS assumes no legal responsibility for the use of this information. ----------

TECHNOLOGY DIVISION
QUALITY PASSED
CHẤT LƯỢNG BAN HÀNH
BỘ PHẬN KỸ THUẬT LUYỆN KIM